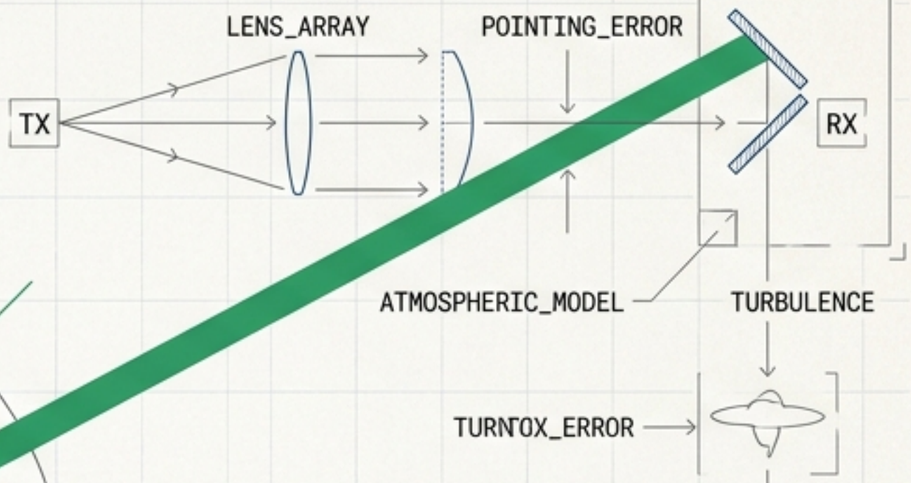
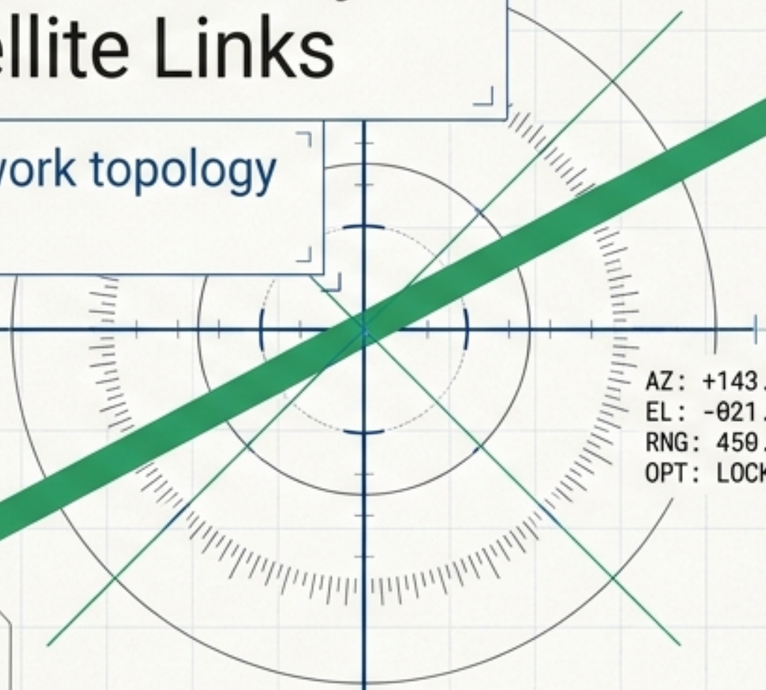
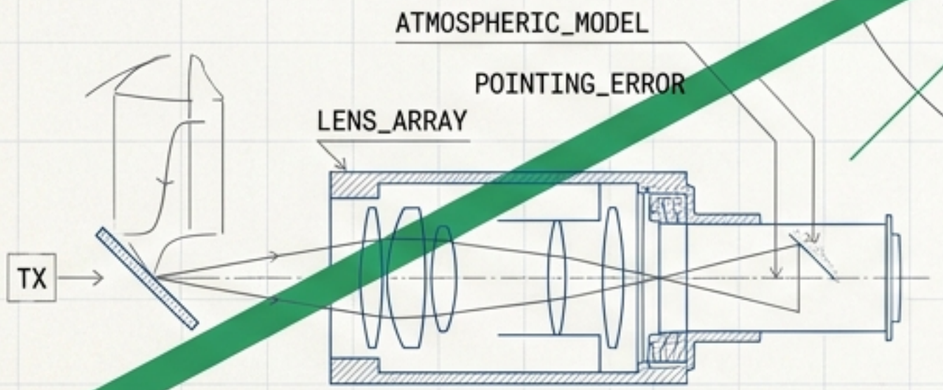
Precision at 430 Terahertz: Channel Modeling & Rate Analysis for Optical Inter-Satellite Links

Balancing beam physics and network topology in Non-Terrestrial Networks







OPTICAL TRANSCEIVER ASSEMBLY CROSS-SECTION



LAT: LEO // TARGET: OISL_NODE

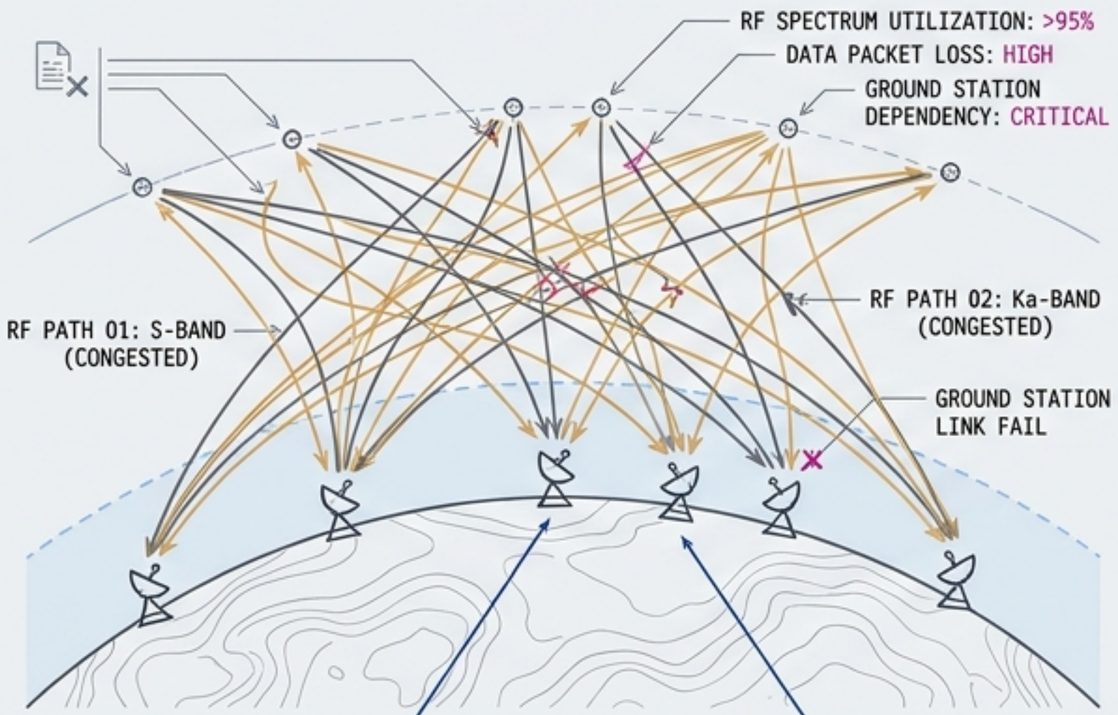
DATA_RATE: 100 Gbps // BER: <1E-9 // OPT_PWR: -12.5 dBm

NotebookLM

NON-TERRESTRIAL NETWORK (NTN) BOTTLENECK: LEAP BEYOND RADIO FREQUENCY

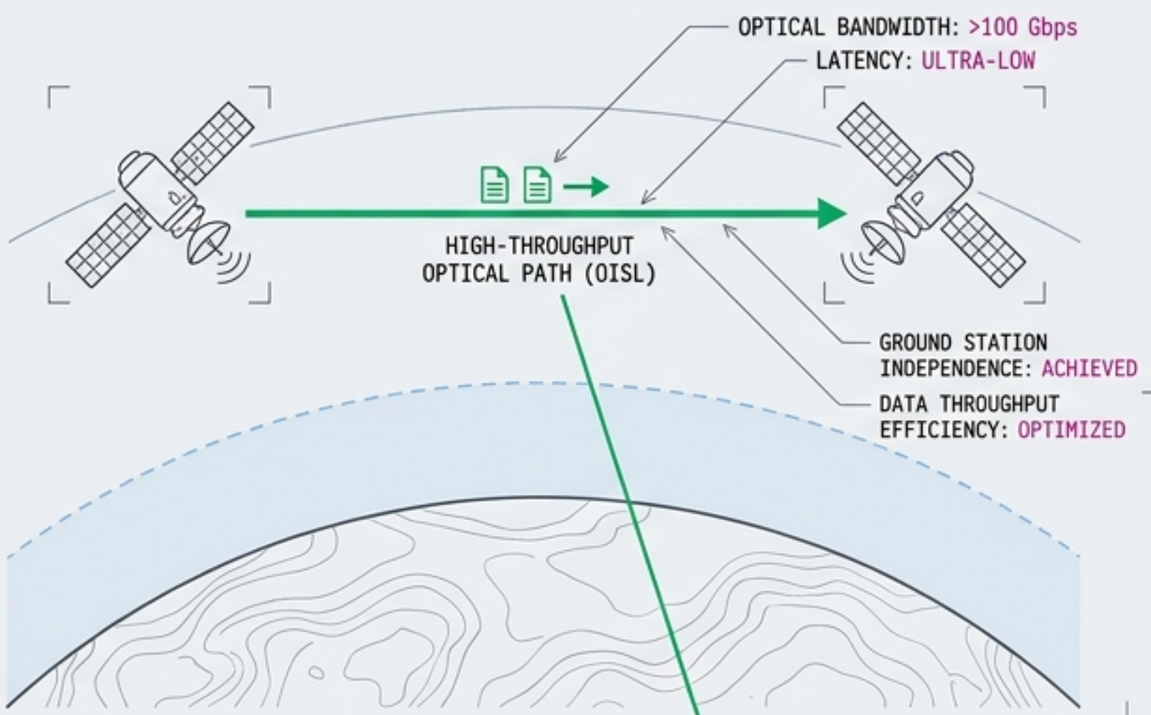
CONGESTED RF ROUTING & GROUND RELIANCE

CONVENTIONAL S/Ka/Ku-BAND LIMITATIONS



OPTIMIZED OISL SOLUTION

INTER-SATELLITE OPTICAL LINKS



[RF LIMIT]

S-band, Ka-band, and Ku-band wireless spectrums are heavily congested.

[GROUND CONSTRAINT]

Satellites currently rely on routing data to designated, fixed ground stations.

[OISL SOLUTION]

Inter-Satellite Links (ISLs) bypass ground stations entirely, requiring a high-throughput medium.

NotebookLM

MOVING TO THE INFRARED SPECTRUM UNLOCKS BANDWIDTH BUT INTRODUCES A NEW PRIMARY OBSTACLE

TRADITIONAL RADIO FREQUENCY
CONVENTIONAL S/Ka/Ku-BAND LIMITATIONS

SPECTRUM RANGE:
S/Ka/Ku bands

BANDWIDTH:
Constrained by congestion

HARDWARE PROFILE (SWaP):
Large antennas,
high power drain



PRIMARY OBSTACLE:
Omnidirectional energy waste

OPTICAL INTER-SATELLITE LINK (OISL)
INTER-SATELLITE OPTICAL LINKS

SPECTRUM RANGE:
Infrared: 300 GHz – 430 THz

[OPTICAL WAVELENGTH: IR]



BANDWIDTH:
Exceptional waveform
encoding capacity

[HIGH-THROUGHPUT DATA]

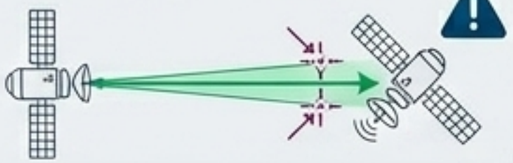


HARDWARE PROFILE (SWaP):
Reduced Size, Weight,
and Power



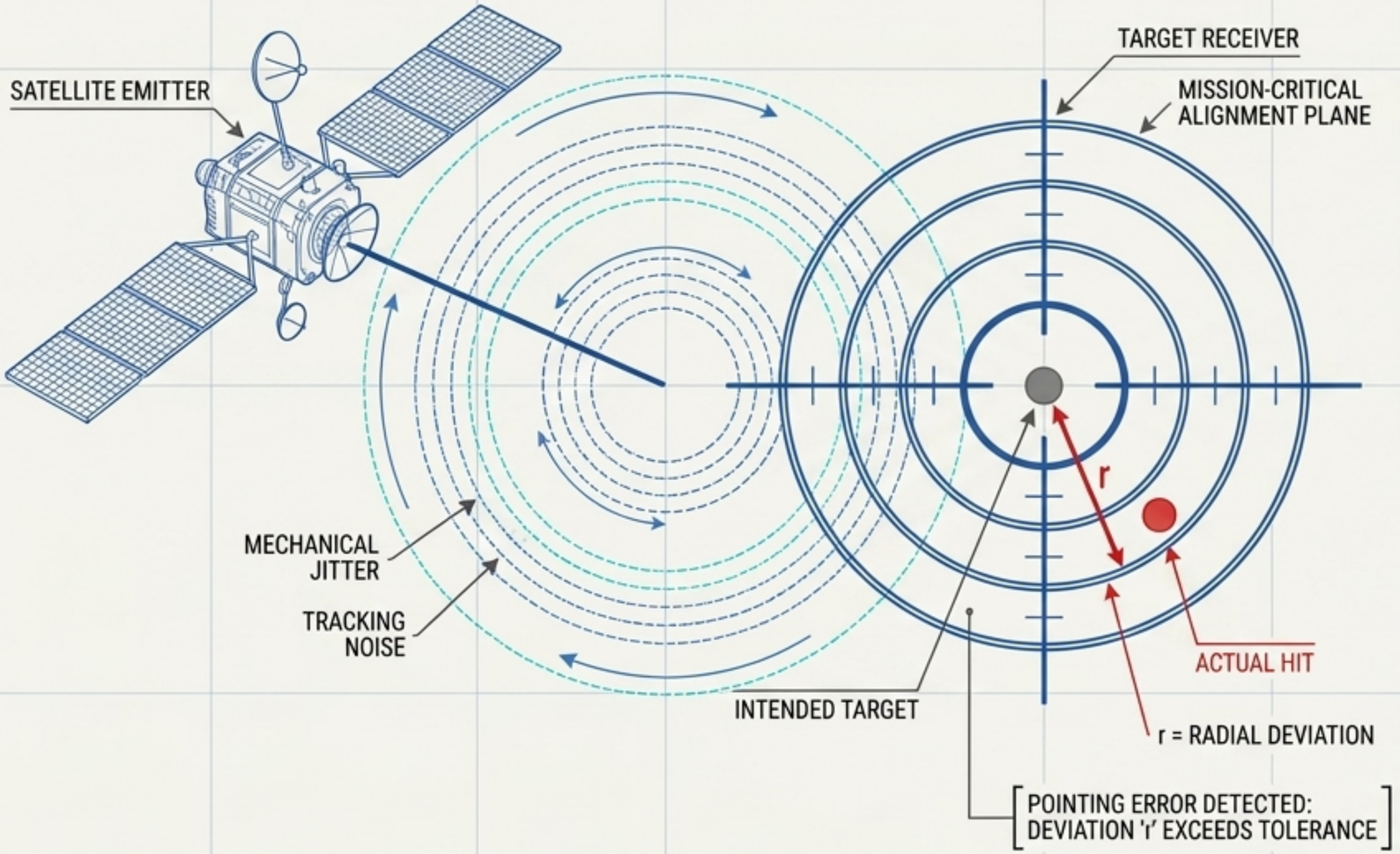
[TARGET ACQUISITION: CRITICAL]

PRIMARY OBSTACLE:
Pointing Errors from
narrow beam alignment



NotebookLM

SATELLITE JITTER AND TRACKING NOISE TURN MILLIMETER MISALIGNMENTS INTO MISSION-CRITICAL POINTING ERRORS



THE SOURCE OF ERROR: MECHANICAL SATELLITE JITTER AND TRACKING NOISE CAUSE THE LASER'S POINTING DIRECTION TO VIBRATE.

TELEMETRY DATA:
PLATFORM STABILITY COMPROMISED.
OSCILLATION DETECTED IN X/Y AXES.
[**CAUSE:** THERMAL FLEX, GYRO DRIFT, ATTITUDE CONTROL SYSTEM NOISE].

RESULT: BEAM PATH VIBRATION >10μrad.

THE MATHEMATICAL REALITY: THE RADIAL DEVIATION DISTANCE (r) FROM THE INTENDED BEAM CENTER FOLLOWS A CONTINUOUS RAYLEIGH DISTRIBUTION.

DISTRIBUTION MODEL: $r \sim \text{Rayleigh}(\sigma_s)$.

[EQUATION: $P(r) = (r / \sigma_s^3) * \exp(-r^2 / 2\sigma_s^2)$].

NOTE: σ_s = STANDARD DEVIATION OF POINTING ERROR IN ONE AXIS.
ERROR MAGNITUDE UNPREDICTABLE, STATISTICALLY MODELED.



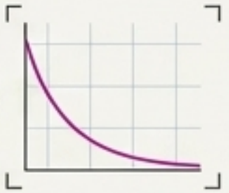
THE CONSEQUENCE: AS DEVIATION INCREASES, RECEIVED SIGNAL POWER DROPS, CONSTRAINED BY POINTING ERROR VARIANCE (σ_s^2).

PERFORMANCE IMPACT:
SIGNAL FADE PROPORTIONAL TO r^2.

POWER LOSS:
$Pr(r) = P_{tx} * G_{tx} * G_{rx} * (\lambda / 4\pi d)^2 * \exp(-G_{tx} * r^2 / 2)$.

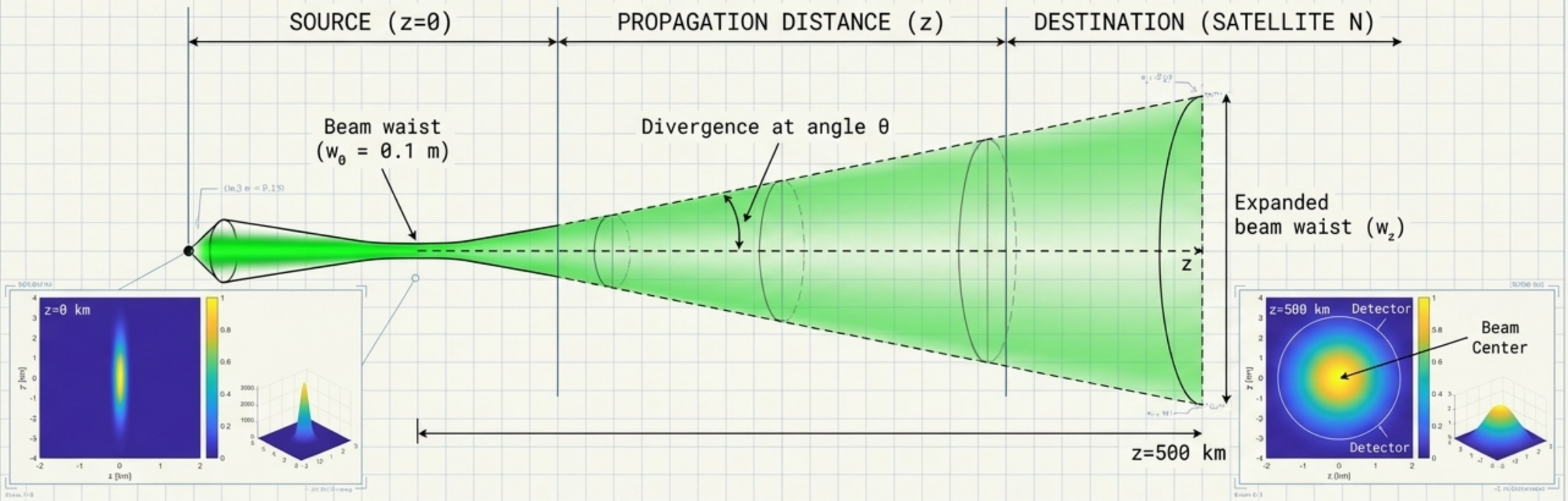
[**WARNING:** RECEIVED SIGNAL POWER DROPS RAPIDLY AS r INCREASES. HIGH VARIANCE (σ_p^2) SIGNIFICANTLY REDUCES LINK AVAILABILITY & DATA THROUGHPUT].

STATUS: LINK MARGIN ERODED.



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GAUSSIAN BEAM EVOLUTION IN FREE-SPACE OPTICS



Source (z=0):

Beam waist ($w_0$0 = 0.1 m).
Maximum power intensity.

Propagation Distance (z):

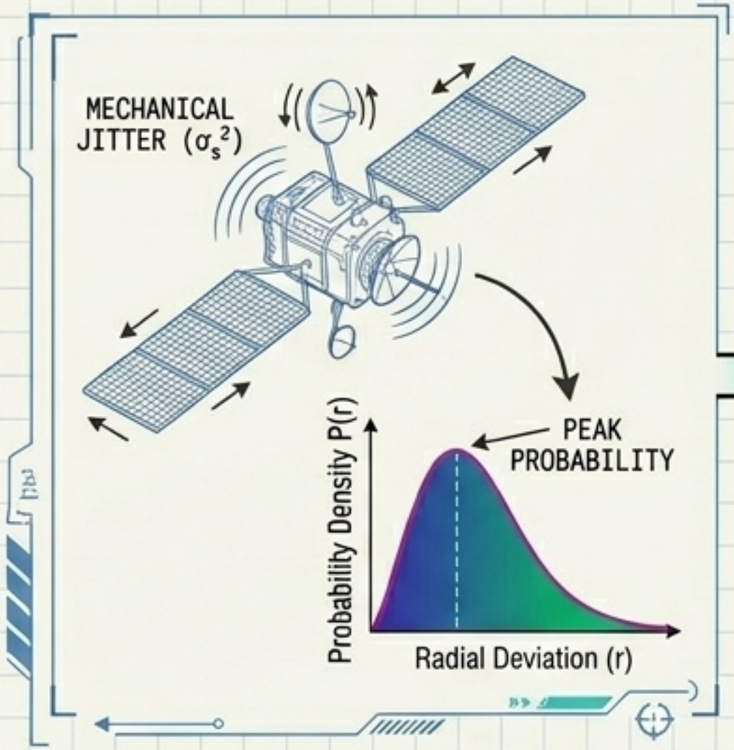
Divergence at angle θ.
Path loss in space vacuum is deterministic (h_{PL} ≈ 1).

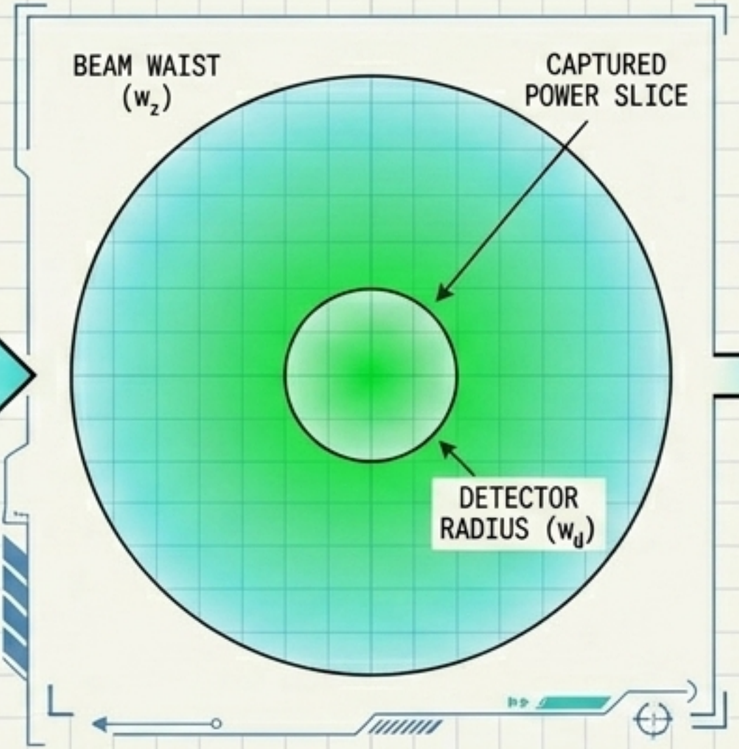
Destination (Satellite N):

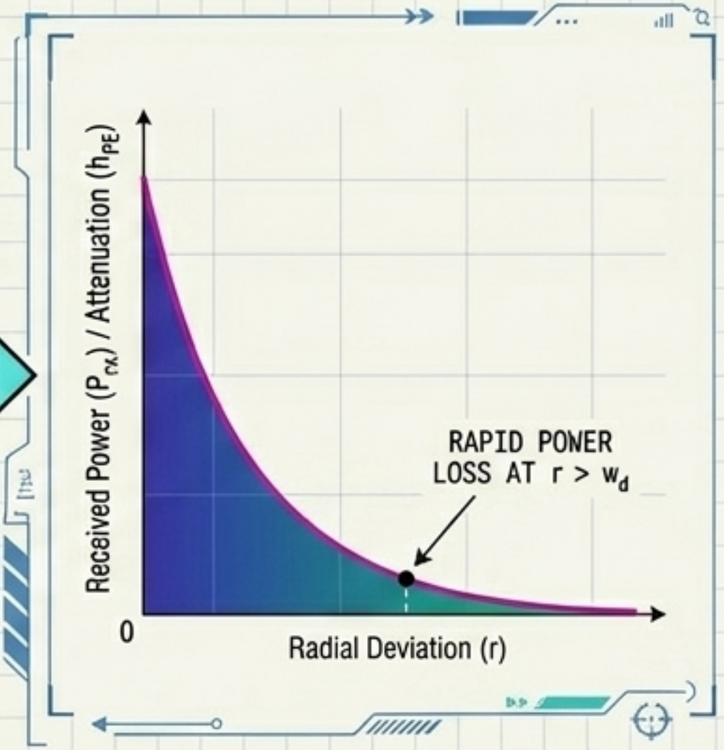
Expanded beam waist (w_z).
Power is diffused across the detector.

NotebookLM

TRANSLATING MECHANICAL VIBRATION INTO CHANNEL STATE ATTENUATION.







Input:

Jitter Variance (σ_s^2) + Gaussian Power Intensity I(r,z)

The Slicer:

Detector Radius (w_d)

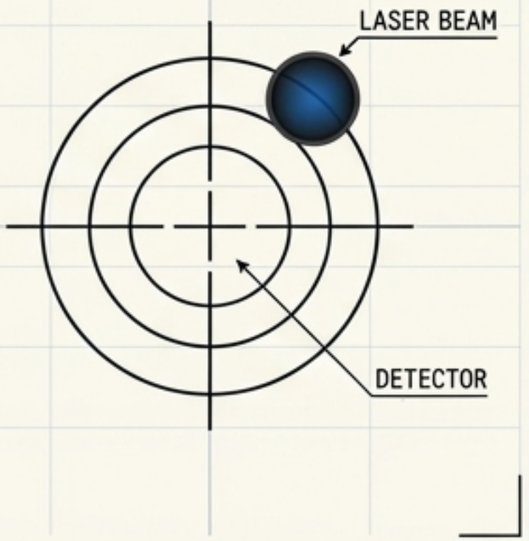
Output:

Channel State Attenuation (h_{PE}). Power captured drops rapidly as radial deviation approaches r_{max}.

NotebookLM

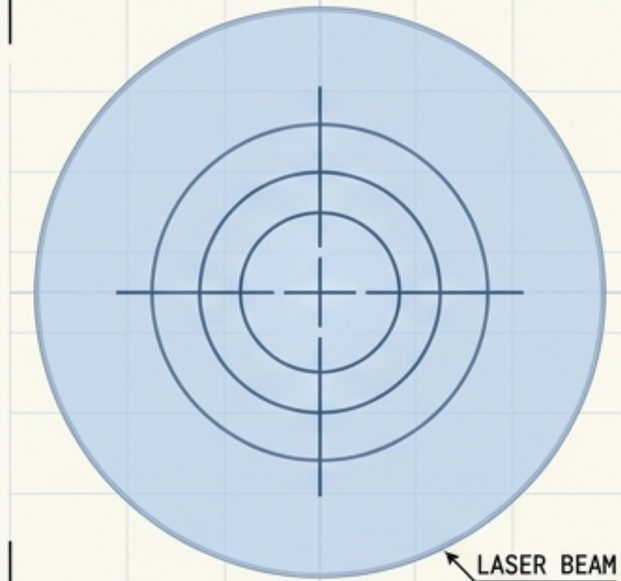
THE PHYSICAL TRADE-OFF: CONCENTRATING BEAM ENERGY VS. BALANCING MISALIGNMENT

STATE 1: HIGH FREQUENCY / SMALL BEAM WAIST



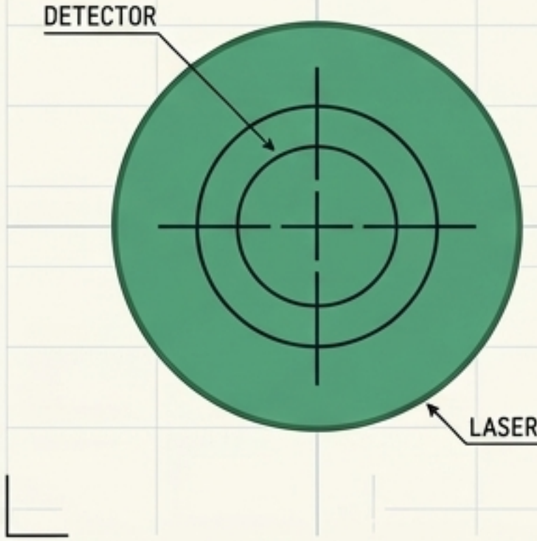
PHYSICS:	MAXIMUM POWER INTENSITY AT THE CENTER.
VULNERABILITY:	HIGHLY VULNERABLE TO JITTER. MINOR POINTING ERRORS PUSH DETECTOR OUT OF THE BEAM.

STATE 2: LOW FREQUENCY / LARGE BEAM WAIST



PHYSICS:	ENERGY HEAVILY DIFFUSED OVER MASSIVE AREA.
VULNERABILITY:	RESILIENT TO JITTER, BUT POWER FALLS BELOW SENSITIVITY THRESHOLD (P_{th}).

STATE 3: THE OPTIMIZED CHANNEL



PHYSICS:	MATHEMATICALLY TUNED TO JITTER VARIANCE.
PERFORMANCE:	ENSURES AVERAGE CHANNEL STATE (h_{th}) CONSISTENTLY EXCEEDS BASELINE.

NotebookLM

THE LONG HAUL: WHY DIRECT SOURCE-TO-DESTINATION LINKS FAIL THE CHANNEL AVERAGE THRESHOLD.



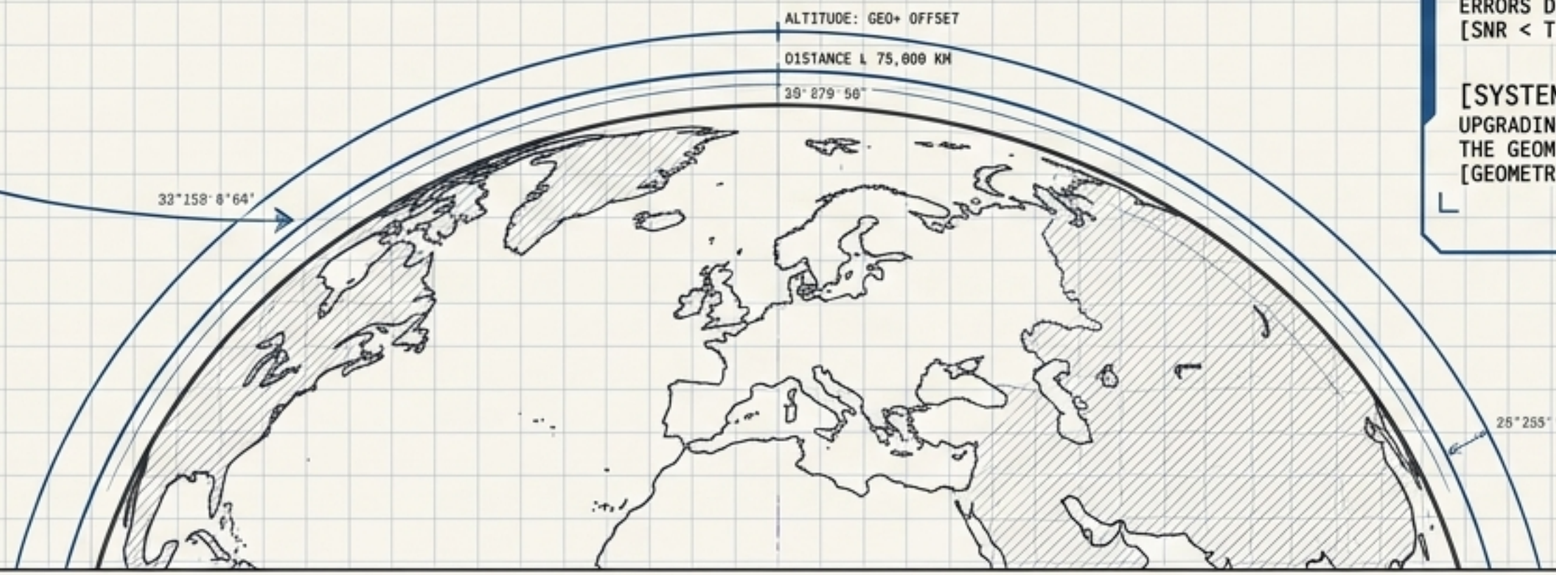
SIGNAL LOSS CRITICAL:
CHANNEL VIABILITY
THRESHOLD BREACHED

POINTING ERROR
SCALE MAXIMIZED

[TRACKING REALITY]
POINTING ERROR VARIANCE SCALES AGGRESSIVELY WITH PROPAGATION DISTANCE.
[$\Delta\theta \propto L^2$] EXPONENTIAL DRIFT OBSERVED.

[SIGNAL DEGRADATION]
EXPANDING BEAM WIDTH AND SEVERE POINTING ERRORS DROP CHANNEL STATE BELOW VIABILITY.
[SNR < THRESHOLD_MIN] MASSIVE DIFFUSION.

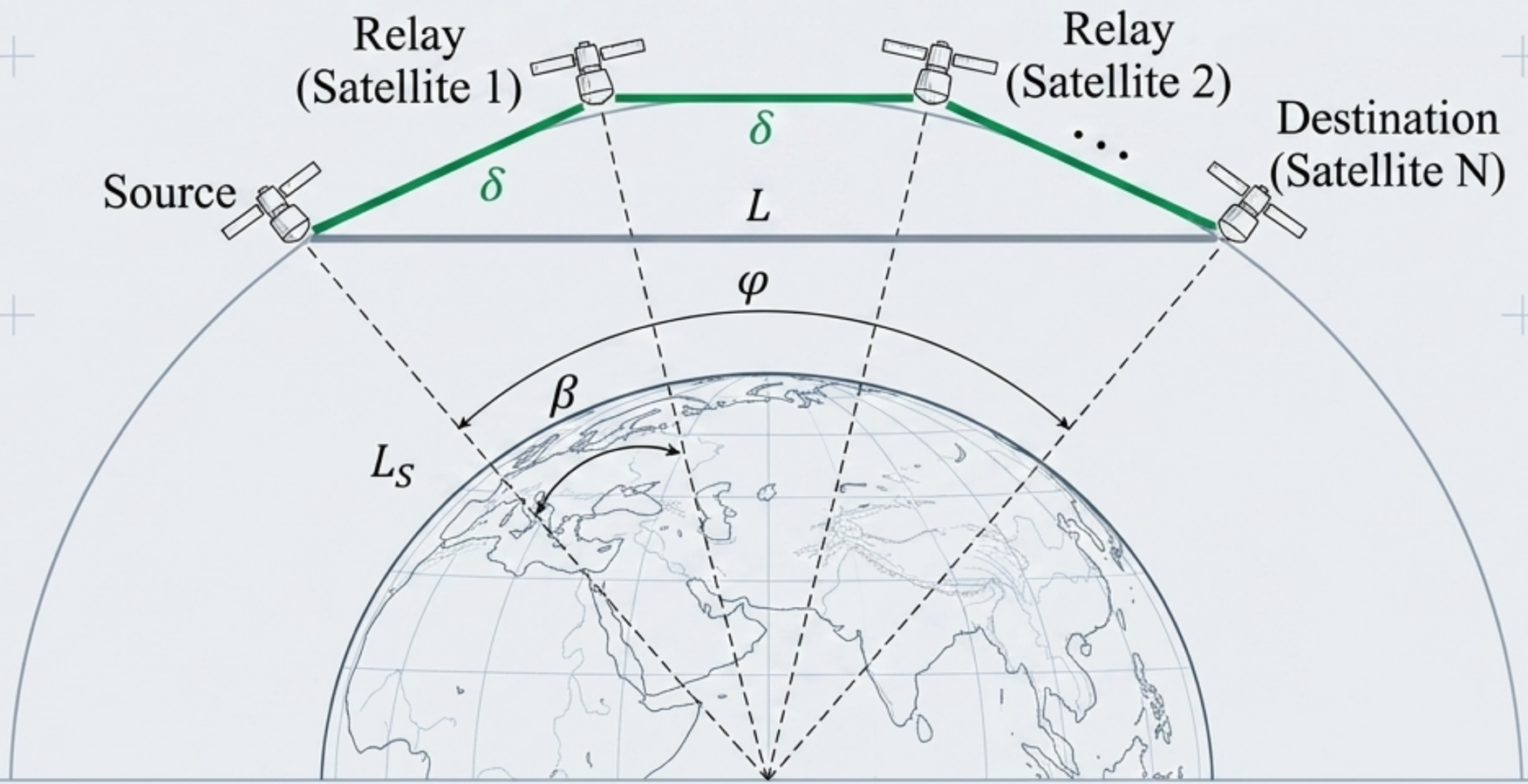
[SYSTEMIC ROADBLOCK]
UPGRADING LASER HARDWARE CANNOT OVERCOME THE GEOMETRY OF A MASSIVE SINGLE HOP.
[GEOMETRIC LIMITS > TECH OPTIMIZATION].



NotebookLM

COOPERATIVE OISL ARCHITECTURE: CIRCUMVENTING PHYSICAL LIMITS WITH MULTI-HOP RELAY NODES.

- **SOURCE TO DESTINATION**: DISTANCE L (e.g., 3000 km). HIGH FAILURE RATE.
- **THE RELAY PATH**: INSERTING N SATELLITES TO BRIDGE THE GAP.
- **THE NEW DISTANCE**: HOP DISTANCE REDUCED TO δ. POINTING ERROR VARIANCE DROPS DRAMATICALLY OVER SHORTER δ DISTANCES.



NotebookLM

THE NETWORK PARADOX: MAXIMIZING PER-LINK THROUGHPUT VERSUS ACCUMULATING TOTAL LATENCY

PER-LINK DATA RATE (R_n)

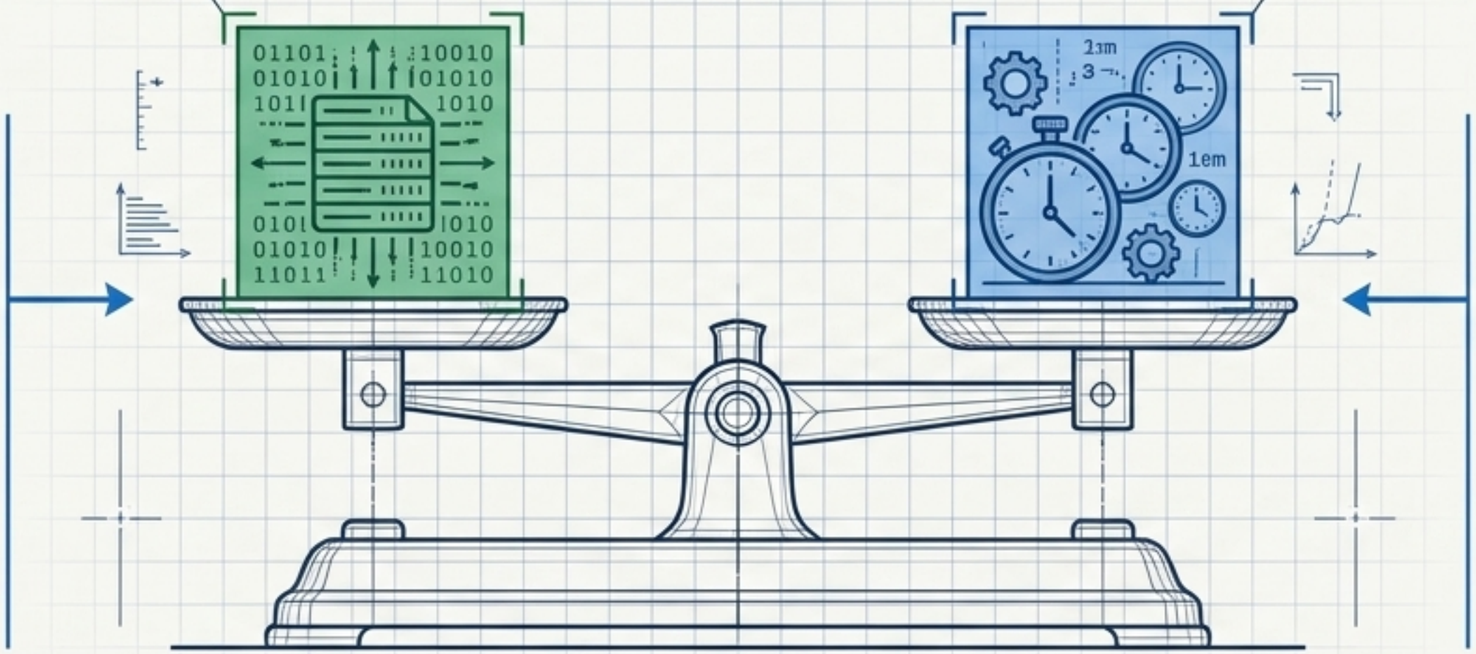
[HIGH THROUGHPUT OPTIMIZATION]

TOTAL CUMULATIVE LATENCY ($\sum T$)

[CRITICAL DELAY ACCUMULATION]

WEIGHT 1: HIGH DATA RATE.

SHRINKING HOP DISTANCE (δ) BY ADDING RELAYS SKYROCKETS THE AVERAGE DATA RATE (R_n) PER LINK.



WEIGHT 2: LATENCY COST.

EVERY ADDED RELAY (N) INTRODUCES PROCESSING DELAYS.

TOO MANY RELAYS VIOLATE THE CRITICAL THRESHOLD (T_{th}).

THE OPTIMIZATION GOAL: MINIMIZE N SUBJECT TO LATENCY CONSTRAINTS

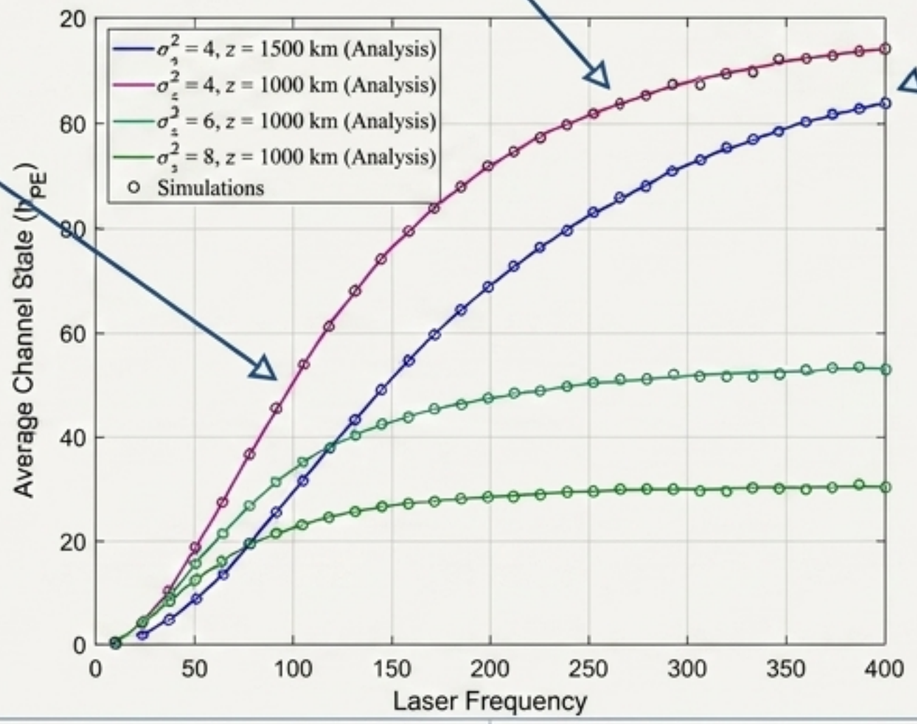
NotebookLM

SIMULATION INSIGHT: HIGHER FREQUENCIES PLATEAU WHEN TRACKING NOISE OVERTAKES BEAM INTENSITY.

THE INITIAL CLIMB:
Channel state (h_{PE}) improves as frequency increases (energy concentrates).

THE PLATEAU:
When jitter is large, higher frequencies fail. The detector physically vibrates out of the ultra-tight beam.

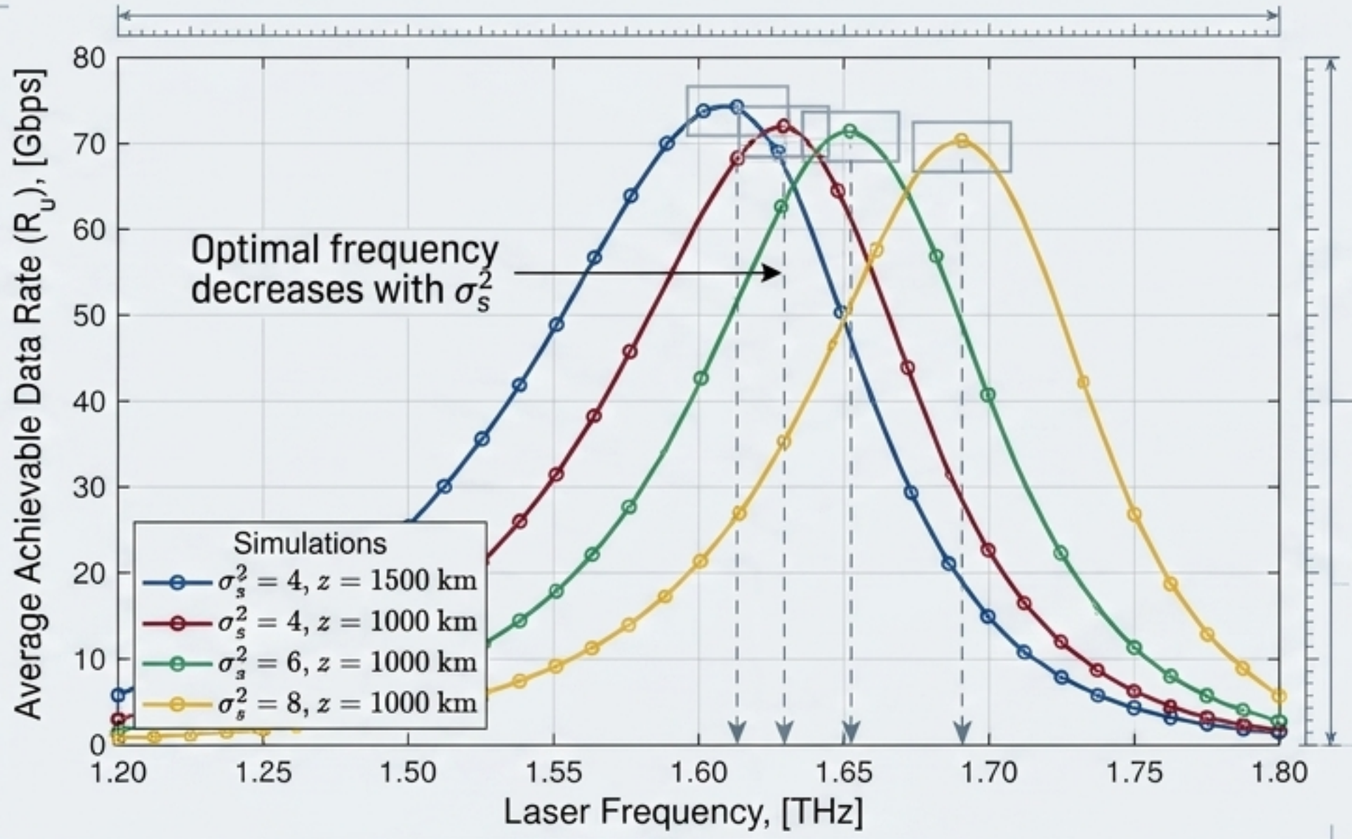
THE DISTANCE PENALTY:
Channel state drops predictably as propagation distance expands from 1000 km to 1500 km.



NotebookLM

THE OPTIMIZATION PEAK: IDENTIFYING THE EXACT LASER FREQUENCY TO MAXIMIZE DATA RATES

THE LEFT SLOPE:
Frequencies are too low; energy is diffused, resulting in lower power at the detector.



THE RIGHT SLOPE:
Frequencies are too high; beams are too narrow, causing severe misalignment penalties from jitter.

THE SHIFTING PEAK:
Optimal laser frequency decreases as pointing error variance increases.

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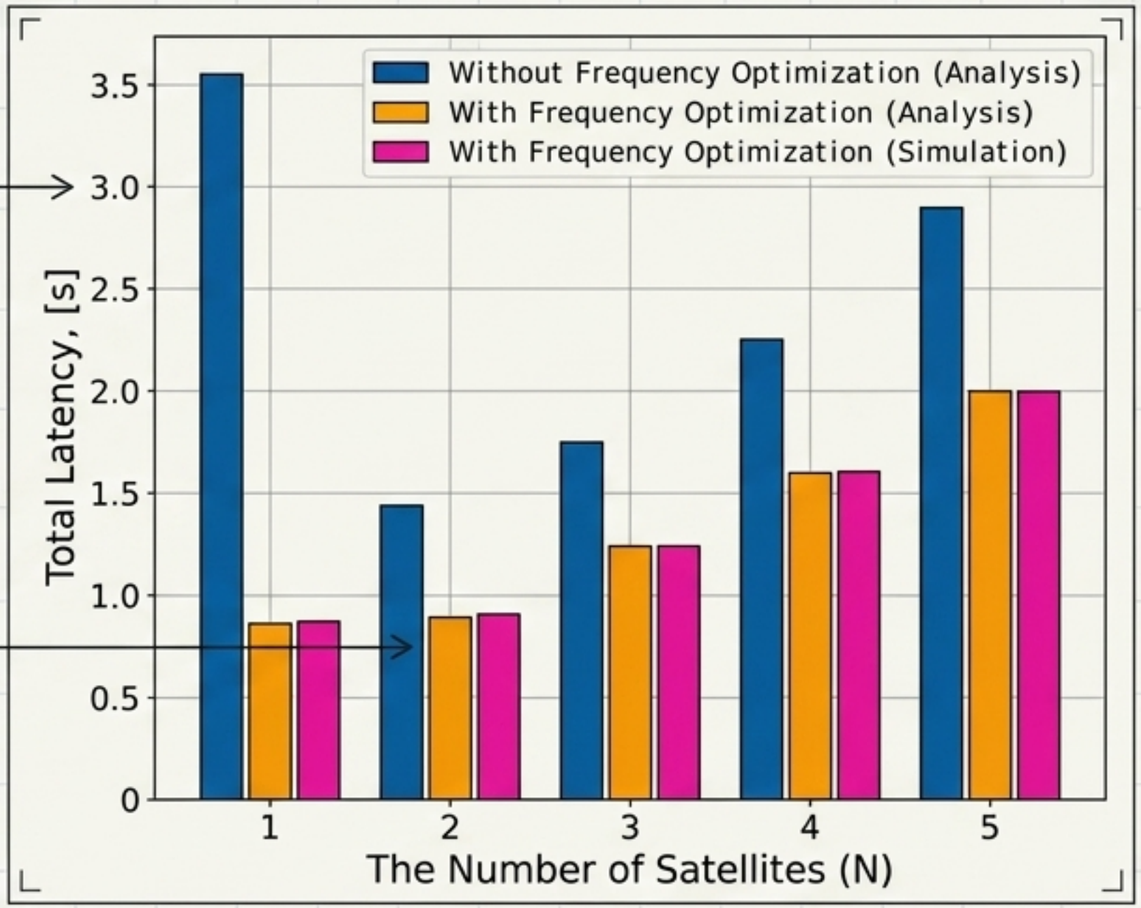
BEATING THE LATENCY CLOCK: SIMULTANEOUS OPTIMIZATION SLASHES TOTAL SYSTEM DELAY

BASELINE: Simulating transmission of 100 Gbits over 3000 km

WITHOUT OPTIMIZATION: Using a default 200 THz frequency results in massive latency spikes.

WITH OPTIMIZATION: Exhaustively searching for perfect relay count (N) and tuned frequency (50-400 THz) guarantees strict T_th adherence.

BASELINE: Simulating transmission of 100 Gbits over 3000 km.



Engineering Blueprint

Targeting Telemetry Document

NotebookLM

THE UNIFIED OISL BLUEPRINT: Network architecture and optical physics cannot be solved in silos.

PHYSICAL LAYER OPTIMIZATION

Beam Waist | Frequency | Jitter Variance

NETWORK LAYER OPTIMIZATION

Relay Count (N) | Hop Distance (δ) | Total Latency

THE UNIFIED BLUEPRINT

Achieving high-throughput optical links requires simultaneous, mathematical harmony. You can only hit the targeted data size within strict latency constraints by concurrently tuning the Terahertz frequency to match mechanical jitter AND deploying the exact mathematical minimum of macroscopic relay nodes.

PROJECT: OISL UNIFICATION

REV 1.0 ENGINEERING SCHEMATIC

NotebookLM

FUTURE HORIZONS: SCALING THE MATHEMATICAL MODEL TO MEGA-CONSTELLATIONS

BASELINE: Expanding framework from point-to-point to global network architecture.

X: -18.6575
Y: 0.857558
Z: -0.8955776



[REV 1.2]
[+-8,-342.5]

[-3,-0.5] [REV 1.2]

MEGA-CONSTELLATIONS:

Applying this channel model to complex, multi-tiered satellite networks.

[+.2.5.0] [ORBITAL_SHIFT_MODEL]

STRICT SINR THRESHOLDS:

Assessing reliability under pre-defined Signal-to-Interference-Plus-Noise Ratio conditions.

[-.1.0.0] [SINR_TARGET > 15dB]

ATMOSPHERIC & ORBITAL PERTURBATIONS:

Expanding the model to account for dynamic orbit shifts and severe angular fluctuations.

Engineering Blueprint
PROJECT: MEGA-CONSTELLATION SCALING

Targeting Telemetry Document
REV 1.2

NotebookLM